Projective Exploration on Individual Stress Levels using Machine Learning

Nagendra Prabhu S
Department Of Computer Science and Engineering Department,
New Horizon College of Engineering
Bangalore,India
snagendraprabhu@gmail.com

Dr. Amarjeet Singh
Department Of Computer Science and Engineering Department,
New Horizon College of Engineering
Bangalore,India

Bhavya B
Department Of Computer Science and Engineering Department,
New Horizon College of Engineering
Bangalore,India
bhavyab0203@gmail.com

K. Sreeja
Department Of Computer Science and Engineering Department,
New Horizon College of Engineering
Bangalore,India
k.sreeja2001@gmail.com

Bhargavi P
Department Of Computer Science and Engineering Department,
New Horizon College of Engineering
Bangalore,India
p.bhargavi28112811@gmail.com

Abstract—**Recently, Stress Prediction in every individual based on their profile and behaviour is a challenging task in the current sector. Current system is a manual process where it is difficult to identify the stress in the college students or employees. There is no automation for any stress prediction. System uses machine learning algorithms or AI algorithms to find out the stress levels, these technologies are used for application development. The purpose of this project is to reduce stress in students and employees. Computer science has come a long way in the last few years. It is massive and multifaceted. It has been used in a variety of applications to meet the basic needs of human society. In the field of healthcare, machine learning has made significant progress. Stress is a deadly disease that kills a large number of people worldwide. We examine how machine learning techniques can assist reduce the risk of stress prediction, which can lead to accidents, in this study.**

Keywords: Machine Learning, ASP.NET, Stress detection, social communication, Labelled and unlabelled data set.

1. INTRODUCTION

In the understudies of students such as graduate, undergraduate, postgraduate, and professional students, machine learning allows for the prediction of the potential of stress prediction. We examine the effectiveness of machine learning algorithms in reducing the risk of stress prediction, leading in early treatment of under studies students, in this work. Nowadays, stress became a major problem and sometimes it may response in a positive or negative way. Now a days the word stress became a major thing which leads to different health problems when it crosses certain level, it complexes day by day.

Now a days both females and males can be introverts. Introverts suffer because they don't understand themselves and they aren't understood when surrounded by extroverts.

So, this project helps them, an introvert may believe that going through an experience in great detail will eventually result in figuring out what went wrong according to those suggestions will be given and help them from stress.

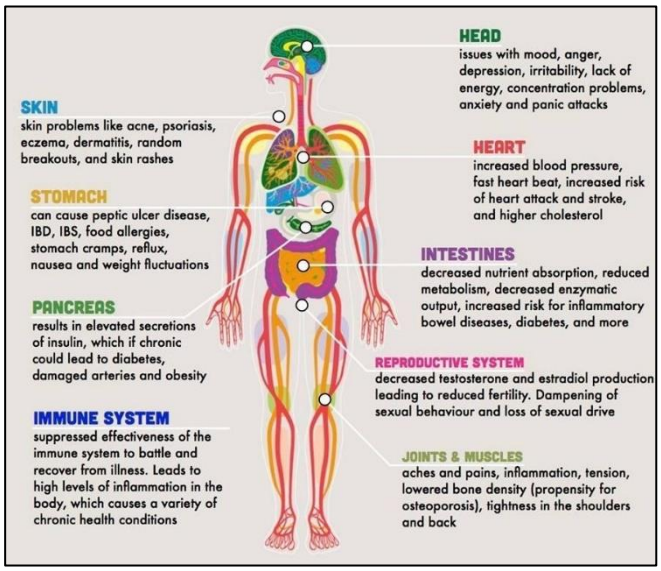

Fig.1.Effects of Stress on the body

In today's culture, the number of people suffering from stress is steadily increasing. The effects of stress on one's entire health are intertwined. According to Hans Selye, stress is "an organism's non-specific response to a demand or a change in its physical circumstances." It might manifest as either eustress or distress. Eustress has a beneficial effect, whereas distress has a negative effect. A pattern formed by cognitive, emotional, behavioral, and physiological responses to negative and toxic characteristics of a stressed organization and environment. It's the outcome of misalignment between people and their jobs, as well as interpersonal friction.

This will assist students, employees, and all individuals in objectively determining the correct vision for themselves. From a psychological standpoint, self-reported measurements are the most common way to assess stress. According to the literature review, data mining tools have not

yet been widely used to analyses stress issues; thereby, we use machine learning techniques to overcome this stress issue.

A. Popular Algorithms

- KNN Classifier: It is a supervised learning algorithm which can be implemented on labelled data. It is used for predicting whether a person needs treatment or not. From the data already known it classifies on how similar its independent variable are to the instance with dependent
- Logistic Regression: The best method that is used for prediction analysis. It is used with binary variable that is dependent on other independent variables. Here, we will use few relevant attributes as independent variable and those having stress and needs treatment that is predicted by the model trained.
- Random Forest Classifier: It is used as the flexible ML algorithm which provides good persistent result even if there is no hyper tuning. They are used as the clusters of decision trees that are working together.

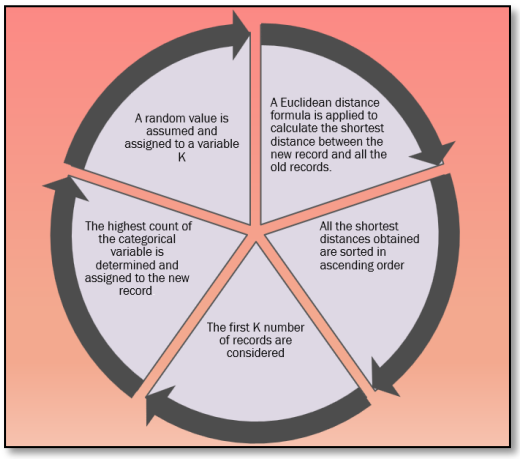

Fig.2. Working of KNN algorithm.

B. Types of Stress

1. Acute Stress:

- Definition: It is a type of short-term stress that comes as a result of daily duties, recent events, or events in the near future. It causes emotional suffering (irritability, anxiety, and anger) as well as medical difficulties.
- Cause:
  - Witnessing a death.
  - Witnessing a severe accident.
  - Exposure to war in the case of either civilians
  - Rape or sexual violence.
  - A physical attack.
- Symptoms:
  - Emotional responsiveness.
  - Difficulty concentrating.
  - Feeling of being disconnected from one's body
  - Sense of numbing
  - Dissociative amnesia.

2. Episodic Acute Stress:

- Definition: It occurs when people are confronted with a series of stressful situations and are aware of impending danger. Aggressive, uptight, and extremely competitive people with a "Form A" mentality are prone to this type of stress. Continuous anxiety, migraines, heart/chest pain, and persistent headaches are all prevalent signs of episodic stress.
- Cause:
  - Diabetes
  - Depression leading to suicidal tendence.
  - Menstrual and skin related problems.
  - Loss of memory
- Symptoms:
  - Panic attacks
  - Irritability
  - Muscle tension

3.Chronic Stress:

- Definition: It is the most dangerous type of stress. It entails constant anxieties that appear to last forever, affecting both the mind and the body.
- Cause:
  - Acne
  - Depression
  - High blood pressure
  - Heart disease
  - Eczema
- Symptoms:
  - Aches and pains
  - Difficulty sleeping
  - Fatigue, Infections
  - Feeling a loss of control

II. LITERATURE SURVEY

Authors **Disha Sharma et al. (2020) [1]** describe that stress among the students they collected the information provided by the students in the form of a quiz or self-rating called a dataset and they used this dataset to make an effective analysis on the stress levels of every student. The information for getting a dataset were collected using both primary as well as secondary fashion. The survey was conducted in the following order such as E-mails, google Drive, survey forms etc., for each questionnaire they provided the students with 5 choices to choose one among them and rate themselves in opinion to their stressful situation. Choice includes:

(A) Never
(B)Almost
(C)often
(D) Very Often
(E) Sometimes

Which had its own rating rank so that it would be easy for detecting the stress faced.

Therefore, the data that was collected is being put together and made use on an pss(perceived stress scale). By using this

source, they discovered the usage of parameters such as True positive rate, false positive rate, recall and f-score.
The formulas for the above is as follows:
TP=True positive, FN=False negative, P=precision.

- TP = TP+ TP/TP +FN.
- FP = FP + FP/TN.
- F-score=2*P*recall/P+ recall.

As per this research we found that they used the technologies such as Naive bayes; Random Forest; Bayes' net; logistic regression.
On comparing, the tool used was weka tool. The overall results on considering various techniques, calculations and comparing we get the following result that is Bayes' net classifier gives the highest accuracy amongst all the other algorithms and provided the final accuracy as 88.5965% as an overall percentage for a particular student's stress level.

Authors **Jarernsri Mitrpanont et al. (2017) [2]** presents a method to create Emotive Device which defines BCI (Brain Computer Interface) technology is used to assess the values that are sensed by sensors which are placed on human scalp. The basic health Questions are a set of 30 questions used to assess stress levels.
The features of this project is:

- Device
- Detect Stress
- Intervention
- results and Observation data report.

Android 4.4, SQLITE, Adobe photoshop, Bluetooth to connect with emotive device, Java programming language these are the following techniques used in this project. Thus, concentration, relaxation and therapy of music interventions are all included in the proportion of stress. To summarise, this paper it found to be a very important tool for assisting individuals in coping with their own stress by utilising the most successful approach known as "Neurofeedback," which gives a perfect option in treating and matching with each and every individual.
Authors **Enrique gorcia et al. (2015) [3]** compares the methods to Auto detect stress among the working environments, they collected the information based on behavioral data which was provided by using the built-in sensors, especially with the use of Samsung galaxy STTI mini smartphones. This dataset was collected in the form of smartphones, written format using applications. From the survey conducted for each questionnaire which resulted in the form of low, medium and high, also it is found that it can identify the rate of stress levels day by day with the help of a smartphone. In accordance with this paper, we found that the technologies used were Naive bayes; and decision trees. By considering various techniques their experiment included 19subjects. Thus, the overall accuracy was found to be 75% which was the first thing on evaluating the potential from mobile phones as a stress detector in the in environment.
Authors **MS. Sumathi M. R and Dr.B. Poorna (2016) [4]** proposed the method to predict attention problems, academic problems, anxiety problems, POD.
In order to predict this, they used Weka tool and artificial intelligent techniques such as Aodesr, multilayer perceptron (MLP), lestar, multi-class classifier (MCC), Ft and Lad tree. This collected information is formed as a dataset with 60 instances.
Therefore, the overall accuracy classifies into different areas such as Kappa statistics, accuracy, and loc area, on comparison among all the attributes and the selected attributes the highest accuracy was found in multilayer perceptron, multi-class classifier and LAD tree.
Authors **Xiyu Liu et al. (2020) [5]** demonstrates model predict the stress among the college students they collected the information provided by the students in the form of a questionnaire which is designed in the form of four dimensions:

- General Demographic
- Epidemic presentation
- Control in the region
- Psychological stress

For each questionnaire they provided the students with 4 choices to choose one among them and rate themselves in opinion to their stressful situation. Choice includes

A. All the time
B. None
C. Occasionally
D. Not clear

They used spss statistics 25.0 software for calculating accuracy. By calculating stress level using above choices it results the emotional display like worry, tension, anxiety, depressed, fear and mood swings. Thus, they can measure of mental health of a student in a covid pandemic through this project.

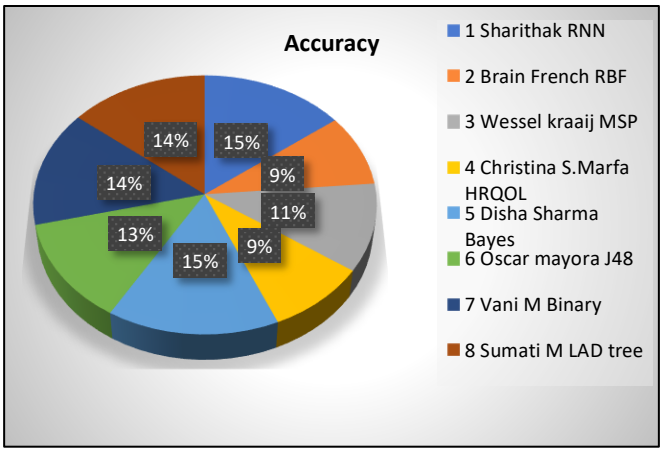

Fig .3 Accuracy differentiation

Authors **Yuan Shi et al. (2019) [6]** described We build a personalized stress detecting model based on SVM, and examined it on the collected data. We examine topics like continuous data, non-intrusive detection of physiological measures, involving data collection, feature extraction, and model construction. The concepts which was ECG, GSR, RIP, EMA, RBF kernels.

- Sessions in the lab study.
- EMA questions and possible answers.
- Features extracted from sensor data
- Numbers of +ve and -ve from subjects

Average and standard errors of precision values
The final result the precision–recall curve of the stress detection model with personalization, segment-based

features, and RBF kernel. Even at 100% recall, the model can still achieve high precision over 50%.

Authors **Saskia Koldijk et al. (2021) [7]** compares the Employees often record their experience of stress at work. This is a literature survey on the paper which is done based on ML, mental stress inference, computer log, detection of facial expressions, gestures and posture.

- There are many options on how to deal with the aspect of time.
- There is debate on whether subjective ratings provide a good ground for truth.

The final performance of Accuracy is 0.8416 which has been reached.

Authors **S. Malfa et al. (2021) [8]**

Suggests the health-related quality of life and psychological stress between public sector and professional groups.

1.Other main characteristics of the participants by this professional group include.

2.Anxiety, depression and HRQoL are the scores (means and SD) by professional category and between- group comparisons.

Authors **Simhadri Naga Mounika et al. (2019) [9]**

The purpose of this paper is that Stress plays a very crucial part in our daily life. This is a literature survey on the paper which is done based on the deep learning concepts those are CNN, RNN, DNN algorithms, FGM for the graph implementation model, SVM for the vector machine, and finally with the neural network. The accuracy is predicted based on the percentage of positive, negative and Neutral.

The highest accuracy is 81.6 for RNN and for CNN is 69.8.

Sno	Authour	Algorithm	Accuracy
1	Sharithak	RNN	81.60%
2	Brain French	RBF	50%
3	Wessel kraaij	MSP	60.80%
4	Christina S.Marfa	HRQOL	49.74%
5	Disha Sharma	Bayes	83.80%
6	Oscar mayora	J48	70.20%
7	Vani M	Binary	81%
8	Sumati M	LAD tree	79%

Fig .4 Algorithm, accuracy differentiation

Authors **Pramod Bobade, et al. (2020) [10]** demonstrates the task was divided into three categories: amusement, baseline, and stress using Dl and ml. The non-stress class was created by combining the amusement and baseline states. Second, the non-stress class was created by combining the amusement and baseline states.In order to predict this, they used WESAD dataset study. This dataset includes Data from numerous physiological modalities such as 3-axis acceleration (ACC), respiration (RESP), and volume blood pulse are also included in the WESAD dataset However during the overall analysis, machine learning techniques yielded accuracy of 81.65 % and 93.20 % for 3-class and binary classification problems and deep learning yielded accuracies up to 84.32 % and 95.21 %, and this was finally predicted as the overall outcome of this paper.

Authors Monisha et al. (2018) **[11]** In this paper the techniques used are mean and standard deviation. according to questionaries they predict yes or no. From using only mean and standard deviation it will not accurate to judge the stress levels. these are the drawbacks.

Authors Sanchitha paul et al. (2016) **[12]** It demonstrated the using the sensory devices like wearable sensors,(ECG), (EEG), and (PPG), KNN Alogrithm, CNN and Machine learning etc. It tries to collect the Physiological signals and sensors for extracting the levels or the dataset values to get the highest accuray for the Feature Extraction.

As it has many various high frequency domain measures it tells that this only the best way to find the solution.

Authors **V.Ranjani,et al. (2020) [13]** In this paper, authors have focused to compare a variety of techniques, approaches and different tools and its impact on the healthcare sector. The goal of a data mining application is to turn that data into facts, numbers, or text which can be processed by a computer into knowledge or information. The main purpose of data mining application in healthcare systems is to develop an automated tool for identifying and disseminating relevant healthcare information. This paper aims to make a detailed study report of different types of data mining applications**[14]** in the healthcare sector and to reduce the complexity of the study of the healthcare data transactions. Also it presents a comparative study of different data mining applications, techniques and different methodologies applied for extracting knowledge from databases generated in the healthcare industry. Finally, the existing data mining techniques with data mining algorithms and its application tools which are more valuable for healthcare services are discussed in detail. The drawbacks for this project is to require huge amount of data and less accurate results.

Authors Alberto de Santos Sierra, Carmen Sánchez Ávila (2018) **[15]** It uses GSR, HR and different logistics. the state of mind of an individual, due to their nonintrusive Ness, specific psychological experiment was designed to induce properly stress on individuals in order to acquire a database for training, validating, and testing the proposed system.

Here it tells that only through 2 ways we can get the highest accuracy and best way to find the solution.

Authors PD purnmasari et al. (2016) **[16]** It is used machines in person to detect the stress with Formations of 60 instances. Using so many instances does give the proper output. It is used to detect only heart rate.

Authors Santhosh Kumar et al. (2019) **[17]** Used ANN concepts. They evaluated the performance of different classifier and compared the result with the three different sets for investigating the effect of role overload on occupational stress. Due to ANN usage too many hidden layers are required for the accuracy performance.

Authors Sami Elzeiny et al. (2018) **[18]** They used few symbols to measure physiological and physical measures such as ECG, EEG, GSR, AGR, IR eye to check heart, brain skin response and gestures for body and eyes through sensors to relive stress. The drawback is due to sensors as it can affect the users.

Authors Parul Agarwalet al. (2020) **[19]** They have used EEG and supervised learning to detect stress and use of k means clustering to group into different sectors of stress. No real time stress detection for this EEG signals.

Authors Jigang Zheng, et al. (2018) **[20]** Data mining from large amounts of data to extract efficient, implicit, potential useful knowledge and information has been successfully

applied. However, the application in the field of education is relatively less. The central work of institutions of higher learning is teaching, the focus is to improve the quality of education, but students are precisely the basis for measuring the quality of teaching. The characteristic of this paper is to apply association rule mining method of university students' grades, previously unknown effects on student achievement factors of mining, to provide some valuable reference for teachers and administrators, to provide the necessary decision support for teaching and student management, to better carry out teaching, in order to improve the quality of teaching in Colleges and universities. The disadvantages are less parameters used and less efficient results.

III. METHODOLOGY

As the starting step for the research the problem is to identify the stress levels in every individual by conducting a survey through google forms or emails.

The collected data is now being understood and trying to fetch the required data for processing here the terms that have most similarities will be grouped based on choices

Operating System	**Windows**
Design Tool	**Visual Studio**
Front End	**ASP.NET 4.0**
Language	**C#**
SQL Server	**SQL Server**
Data Access Technology	**ADO.NET**

Fig .5 Implementation requirements.

A model will be built using machine learning techniques and data accessing techniques such as ADO.NET. These will be further classified with the help of several algorithmic techniques, like for this approach we are trying to implement using KNN-classification algorithm and further is classified into sub divisions such as supervised learning and unsupervised.

Technology Details:

1.Introduction to ASP.NET:

ASP.NET is a unified Web development platform that provides the services necessary to build enterprise-class Web applications. While ASP.NET is largely syntax compatible with Active Server Pages (ASP), it provides a new programming model and infrastructure that allows to create a powerful new class of applications. ASP.NET is part of the .NET Framework and allows to take full advantage of the features of the common language runtime, such as type safety, inheritance, language interoperability, and versioning.

2.Overview of .NET:

The common language runtime is the foundation of the .NET Framework. The runtime can be thought as an agent that manages code at execution time, providing core services such as memory management, thread management, and remoting, while also enforcing strict type safety and other forms of code accuracy that ensure security and robustness. In fact, the concept of code management is a fundamental principle of runtime. Code that targets the runtime is known as managed code, while code that does not target the runtime is known as unmanaged code.

The class library, the other main component of the .NET Framework, is a comprehensive, object-oriented collection of reusable types that you can use to develop applications ranging from traditional command-line or graphical user interface (GUI) applications to applications based on the latest innovations provided by ASP.NET, such as Web Forms and XML Web services.

3.Introduction to C#:

C# (pronounced as ‘C Sharp’) is a new computer-programming language developed by Microsoft Corporation, USA. C# is a fully object-oriented language like Java and is the first Component-oriented language. It has been designed to support the key features of .NET Framework, the new development platform of Microsoft for building component-based software solutions. It is a simple, efficient, productive and type-safe language derived from the popular C and C++ languages. Although it belongs to the family of C/C++, it is a purely object- oriented, modern language suitable for developing Web-based applications.

4.ADO.NET-Database Connectivity:

Most applications need data access at one point of time making it a crucial component when working with applications. Data access is making the application interact with a database, where all the data is stored. Different applications have different requirements for database access. ASP.NET uses ADO .NET (Active X Data Object) as its data access and manipulation protocol which also enables to work with data on the Internet.

5.Introduction to SQL server:

Microsoft SQL Server is a full-featured relational database management system (RDBMS) that offers a variety of administrative tools to ease the burdens of database development, maintenance and administration. Most frequently used tools include, Enterprise Manager, Query Analyzer, SQL Profiler, Service Manager, Data Transformation Services and Books Online.

Enterprise Manager is the main administrative console for SQL Server installations. It provides the client with a graphical "birds-eye" view of all of the SQL Server installations on client’s network. High-level administrative functions can be performed that affect one or more servers, schedule common maintenance tasks or create and modify the structure of individual databases.

Dataset Implementation:

Attributes are identified by the dataset and listed in the form of a table. Where all attributes are of nominal types whereas some attributes which is relevant for classification and prediction for the problem, here the dataset will be pre-processed by eliminating irrelevant and redundant attributes using three tier architecture they are data layer, business layer and presentation layer. For accessing the data, the technology that will be used is ado.net, for the same asp.net would be used as a programming script that is like a front end to develop the code.

SL No	Factors	Sub Factors	Encoded Values
1	Gender		• 1-Male • 2-Female
2	Financial Issues	•Repaying of loans •Deadline of fee payment •Payment For Paying Guest(PG) or Hostel	• 0-No • 1-Yes
3	Family Issues	•Parental Expectations •Being bullied by siblings •Divorce of Parents •Negligence of Children •Poor Communication and misunderstandings	• 0-No • 1-Yes
4	Study Hours		• 1 • 2 • 3.....etc
5	Teaching Method		• 0-Fair • 1-Not Good
6	Health Issues	•Malnutrition •Sinus or Migraine •Insomania(Sleep deprivation	• 0-No • 1-Yes
7	Partiality Fix		• 0-No • 1-Yes
8	Exam Schedule		• 1-Monthly • 2-Half • 3-yearly • 4-Annual • 5-Slip Test
9	Friends Issues	•Conflicts •Comparison between them •Jealousy •Mistrust •Betrayal	• 0-No • 1-Yes
10	Pressure	•Mental disparity and anxiety because of covid. •Students and their close ones getting infected by covid •No enough infrastructure for Online Classes in Pandamic	• 0-No • 1-Yes
11	Regular	1.Hostler •Home Sick •Lack of nutritious food •Lack of Resources 2.Localite •Time management •Transportation Problem	• 0-No • 1-Yes
12	Interaction		• 1-Excellent • 2-good • 3-Average • 4-Poor

Fig.6 Parameters used to implement dataset

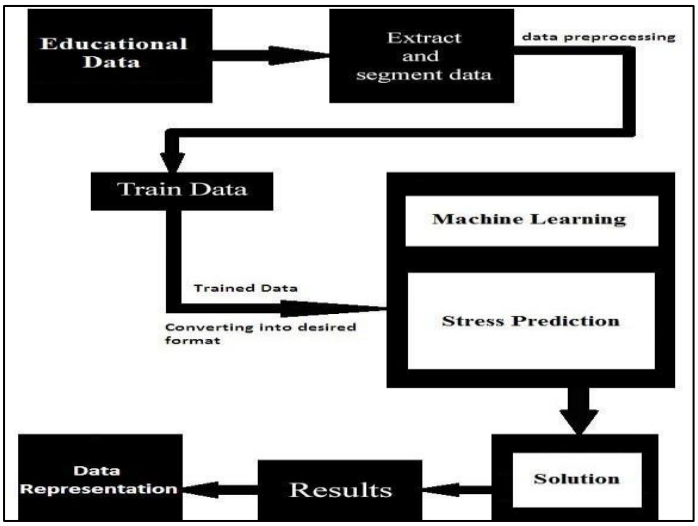

Fig .7 System architecture

In the Existing system individuals are facing so many mental health problems such as depression, pressure, stress, interpersonal sensitivity, fear and nervousness.
Though many industries and corporations provide mental health related schemes and try to ease the workplace atmosphere, the issue is far from control.
Limitations for the existing system
• No Automation for stress prediction
• Less user satisfaction
• Time consuming, Less Reliable, Less Efficient
• Manual Process

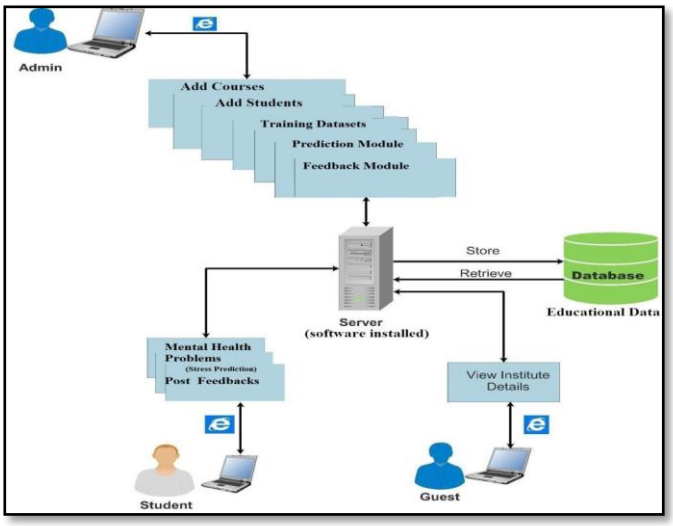

Fig .8 Overview of functionalities of the module

Hence, this proposed project helps in predicting the stress levels due to various parameters considered through a survey and it also figures out whether the individuals are stress free or stressful and also displays the range of their stress levels along with the necessary suggestion in other words, say guidance or precautions.

IV. CONCLUSION

- Proposed system is a real time application.
- Proposed system gives better decisions and also improvises the business.
- Proposed system makes use of data science technique "classification rules" for predicting stress among individuals.
- According to the literature review, data mining tools have not yet been widely used to analyses stress issues thereby, we use machine learning techniques to overcome this stress issue.
- The student can incorporate the solution and work towards maintaining his or her mental equilibrium.
- So, this project helps them, an introvert may believe that going through an experience in great detail will eventually result in figuring out what went wrong according to those suggestions will be given and help them from stress.

- Using machine learning techniques, we predict student stress level and also proposed system will give suggestions based on the stress levels of student.
- Machine learning methods for prediction of stress and mental health condition gives significant results and can be explored further, meeting the objectives of this paper.

V. FUTURE WORK

- With changing lifestyle and work culture, there is an increase in the risk of stress among the college students. System finds factors that strongly determine the stress levels.
- Stress was identified based on gender, family history and availability of health benefits in this sector.
- By identifying the stress of among individuals, system comes up with some approaches to reduce stress and create a much comfortable zone.
- System uses many parameters such as gender, age, family history, provided health benefits, sharing about illness, tech company, tech role, acquiring leave etc.
- System uses machine learning algorithms or AI algorithms to find stress of a student.
- System can be developed as a real time application which is useful for colleges. As Visual Studio and SQL Server is more supportive with real time applications, these technologies are used for application dev.

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